AYE
TECH HUB

• MEP ENGINEERING • LESSON 04

HVAC Fundamentals for MEP

Cooling Loads · System Selection · Zone Control · Integration

Lesson 04 of 30 | MEP Engineering Program

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- HVAC system types for MEP — central vs distributed vs VRF
- Cooling and heating load calculation principles for buildings
- Zone control: how VAV, FCU and VRF serve different building areas
- HVAC-electrical integration: kW demand, power factor, control panels
- Reading and producing an HVAC schedule on MEP drawings

HVAC in the MEP Context

HVAC is the "M" — the Mechanical — in MEP. For an MEP engineer, HVAC is not just about comfort. It is about **coordinating the largest mechanical system in a building with the electrical supply, the structural constraints, and the plumbing distribution**. The HVAC designer must understand not only how to size equipment but how it integrates with every other system in the building.

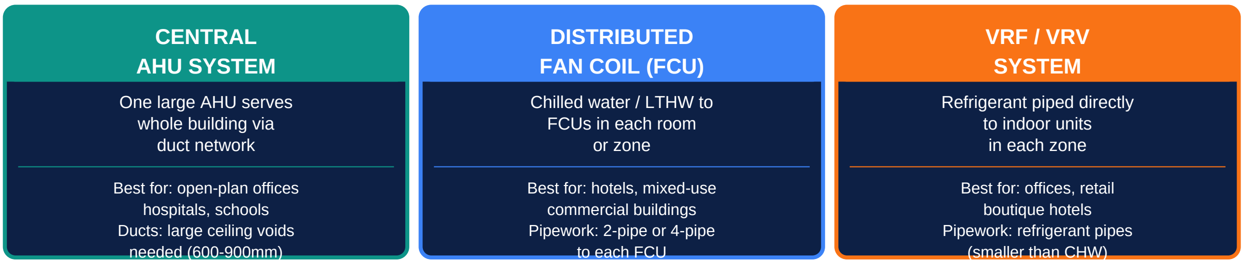

Figure 1 — Three HVAC System Types for MEP Buildings

MEP Consideration	Central AHU	Fan Coil (FCU)	VRF System
Ceiling void required	600–900mm for main ducts + 300mm branch ducts	350–500mm for CHW pipework + slim FCU cassette	200–300mm for refrigerant pipes only
Plant room requirement	Large AHU plant room (≥ 5% of GFA)	Chiller + cooling tower plant room (3–6% GFA)	Outdoor condensing units on roof or at grade (1–2% GFA)
Electrical connection per system	1 × large 3-phase supply for AHU + controls and VAV actuators	3-phase for chillers + pumps; 1-phase per FCU	3-phase per outdoor unit; 1-phase per indoor unit
BMS integration complexity	High — VAV dampers, AHU sequencing, zone sensors	Medium — CHW valves, FCU speed control per zone	Medium — VRF controller integration to BMS via gateway
Fresh air / ventilation	Handled within AHU — HRV, filtration, heating/cooling	Separate MVHR or dedicated OAHU for fresh air	Separate dedicated OA unit required — VRF cools/heats only

The MEP engineer's HVAC role: In most projects, a specialist HVAC consultant designs the system in detail. The MEP coordinator's role is to: verify spatial requirements are met (ceiling void, plant room), coordinate electrical supply to HVAC equipment, ensure fire dampers are in duct penetrations, verify BMS integration, and check clash-free routing of ducts/pipes with structural and other MEP services.

Cooling and Heating Load Fundamentals

Before any HVAC system can be specified, the **peak cooling and heating loads** of each space must be calculated. The load determines the capacity of every piece of equipment — AHU, chiller, boiler, FCU — in the system. An undersized system fails to maintain comfort. An oversized system cycles inefficiently and costs more to install.

Typical Commercial Building Cooling Load Breakdown

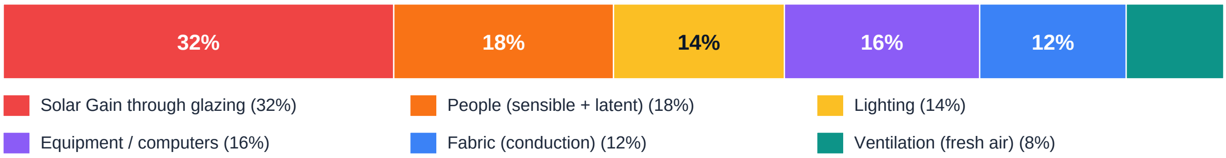

Figure 2 — Cooling Load Components in a Typical Commercial Building

Cooling Load Calculation — Key Parameters

Load Component	Formula / Method	Typical Value
Solar gain through glazing	$Q = A_{glass} \times SHGC \times SC \times I_{solar}$ Where I_{solar} = peak solar irradiance for orientation	South-facing: 500–800 W/m^2 glass North-facing: 100–200 W/m^2 glass
People — sensible heat	$Q_s = N_{people} \times$ sensible heat gain per person	Office: 70W/person sensible Restaurant: 90W/person sensible
People — latent heat (moisture)	$Q_l = N_{people} \times$ latent heat gain per person	Office: 45W/person latent Gym: 200W/person latent
Lighting	Q = Total installed lighting power (W) × diversity	10–12 W/m^2 office; 15-25 W/m^2 retail Diversity factor 0.7–0.9
Equipment / small power	Q = Total small power (W) × diversity × fraction heat radiated	Office: 15–25 W/m^2; Server room: 500–2000 W/rack
Fabric conduction	$Q = U \times A \times (T_{outside} - T_{inside})$ Summer peak ΔT for each construction element	Well-insulated wall U=0.25: 5–8 W/m^2 Single glazing: 20–30 W/m^2
Fresh air / ventilation load	$Q = 1.2 \times V_{oa}$ (m^3/s) $\times (h_{outside} - h_{inside})$ Psychrometric calculation	Hot humid climate: can be 30–50% of total load

Design Conditions and Rules of Thumb

Space Type	Cooling Load (W/m^2)	Heating Load (W/m^2)	Occupancy (m^2/person)
Open-plan office	80–120	30–60	8–12
Meeting / conference	120–180	40–70	2–4
Retail (supermarket)	150–250	40–80	5–10
Hotel bedroom	60–100	25–45	—
Restaurant / dining	150–250	50–80	1.5–2
Hospital ward	80–120	30–60	8–12 (bed + care area)
Server room / IT	500–2000+	—	High density — bespoke
Warehouse / logistics	20–50	15–30	Low occupancy

Zone Control, HVAC-Electrical Integration and Schedules

Different spaces in a building have different thermal needs, different occupancy patterns, and different solar exposures. Zone control ensures that each area receives the right amount of conditioning at the right time — without wasting energy conditioning spaces that are unoccupied.

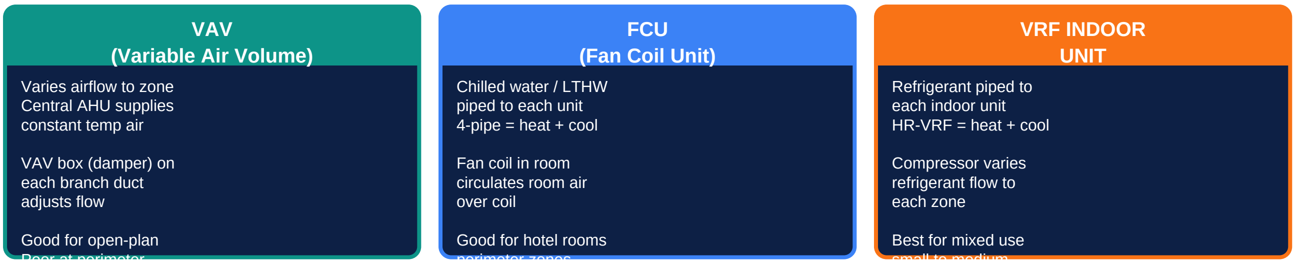

Figure 3 — Zone Control Systems: VAV / FCU / VRF Comparison

HVAC-Electrical Integration — What the MEP Engineer Coordinates

HVAC Equipment	Electrical Connection Required	Key MEP Coordination
Air Handling Unit (AHU)	3-phase 400V; typical 5–50 kW per AHU Starter panel or VFD in plant room	Isolator, supply cable, earthing; controls interface to BMS; fire alarm interlock for smoke detection
Chiller (air-cooled)	3-phase 400V; 10–500+ kW Direct-on-line or soft-starter	Heavy cable (large CSA); power factor correction; co-ordinate roof location with structural
Cooling Tower	3-phase; 2–30 kW fan motor Plus water treatment dosing pump	Roof coordination; water treatment electrical supply; winterisation heater supply
Fan Coil Unit (FCU)	1-phase 230V; 100–500W each Individual isolator per unit	Local switched fused spur; controls cable to room thermostat and BMS
VRF Outdoor Unit	3-phase 400V; 8–40 kW per outdoor unit Dedicated circuit per unit	Cable route from electrical riser to roof; refrigerant pipe route coordination
BMS Control Panels	Small power for DDC controllers Low voltage controls wiring to all sensors	Control cable routes throughout building; interface to electrical distribution boards

The HVAC Equipment Schedule on MEP Drawings

Schedule Column	What It Shows	Source / How Determined
Equipment Tag	Unique reference (e.g. AHU-01, FCU-B2-03)	Assigned on MEP floor plan drawing
Location / Floor	Which floor or plant room	From coordinated MEP floor plans
Duty (kW or m³/h)	Cooling / heating output or airflow	From load calculations — overseen by MEP consultant
Electrical Supply	Phase, voltage, kW, starting method	From manufacturer data for selected equipment
Refrigerant / CHW connection	Pipe size and medium (CHW, refrigerant, LTHW)	From pipework schedule on MEP drawings

Controls reference	BMS point list, thermostat reference	From BMS controls schedule — coordinated with controls engineer
Manufacturer / Model	Approved product reference	From consultant's equipment specification or approved alternatives

What comes next — MEP-005: Heating Systems and Boilers — the mechanical heating side of MEP: hot water boiler systems, low-temperature heating, heat pumps as MEP plant, underfloor heating integration, and the heating pipework distribution strategy for buildings.